SkillUp

Final Project: Comprehensive AI-Enhanced Business Analysis Portfolio

Estimated time: 90 minutes

Introduction

Welcome to the final project, where you will simulate a professional consulting engagement for an online bank that is preparing to launch a digital loan pre-approval and onboarding system. Using generative AI tools such as ChatGPT, Claude, or Copilot, you will plan, model, and document the solution. The goal is to apply what you have learned about AI-assisted business analysis and systems design to produce a set of high-quality artifacts.

By completing this project, you'll demonstrate your mastery of AI-enhanced analysis and create valuable resources you can immediately use in your professional practice.

Learning objectives

- Apply prompt engineering and generative AI tools to gather business requirements and produce accurate documentation, such as user stories, use cases, and acceptance criteria, from unstructured inputs.
- Generate, critique, and improve workflows by producing BPMN diagrams, process flowcharts, and customer journey maps using generative AI.
- Create structured business analysis outputs and communication assets, such as business rules, dashboards, executive briefs, and stakeholder-specific summaries.
- Integrate generative AI with business analysis platforms, such as Jira, Confluence, Miro, and Power BI, to improve deliverable quality and efficiency.

Project scenario

You're working for an online bank that wants to streamline its loan pre-approval process to give customers an automated digital onboarding experience. The system must collect KYC data, perform a soft credit pull, generate a pre-approval decision within minutes, and route complex cases to human agents. The process should function across web and mobile platforms, log customer consent, and comply with privacy and audit standards.

The project assumes that future enhancements might include multilingual support, document upload, and customer experience features, but these are out of scope for the current MVP.

You'll perform all tasks using various AI tools. Once you complete the tasks, you'll save them as DOCX or PDF files, and then submit them as a comprehensive project in the final project submission assignment. See the Project Overview reading for task templates.

Project tasks

This final project consists of six tasks:

Task 1: Use generative AI for discovery and requirements

- Create a interview plan
- Simulate stakeholder interviews

Task 2: Create a process model and architecture

- Create a to-be BPMN process model
- Identify key workflow steps
- Use AI to generate a BPMN
- Design a customer journey map

Task 3: Define the rules, data, and insights

- Define business rules
- Define data entities and attributes

Task 4: Define quality, deployment, and compliance

- Create test cases
- Build a User Acceptance Testing (UAT) plan
- Create a deployment and rollback readiness checklist
- Document compliance, performance, and scalability

Task 5: Create communication and platform stubs

- Write an executive brief
- Create a stakeholder map and engagement plan
- Generate a multilingual executive brief
- Create platform implementation stubs

Task 6: Reflect on your AI usage

- Summarize your AI workflow
- Document ethical considerations

After completing the tasks, you will submit each task document separately in the project submission assignment.

Templates

Use these template structures and save your responses. Save as a DOCX or a PDF file.

- Task1_Discovery_and_Requirements_Template
- Task2_Process_and_Architecture_Template
- Task3_Rules_Data_Insights_Template
- Task4_Quality_Deployment_Compliance_Template
- Task5_Communication_Platform_Stubs_Template
- Task6_AI_Usage_Reflection_Template

Prerequisites

To complete this project successfully, you should have:

- A laptop or desktop computer to conduct the final project
- Stable internet connection
- Access to a word processing software, such as Google Docs or Microsoft Word (to save DOCX or PDF files)

Let's get started!

Task 1: Use generative AI for discovery and requirements

In this task, you will simulate the discovery phase of a system design project. You will use generative AI to create interview questions, simulate interviews, and identify requirements for the onboarding system.

Instructions

Part 1: Create an interview plan

You will design and organize your discovery interviews. You will create 10 role-based stakeholder questions designed to capture business objectives, compliance expectations, and operational needs.

1. Identify stakeholder roles
   You will focus on three key stakeholder groups:
   • Product team – defines business goals and features
   • Compliance team – ensures legal and regulatory alignment
   • Operations team – manages workflow efficiency and manual review processes

2. Define interview objectives
   Before generating questions, clarify what you need to learn from interviews. At a minimum, your interview objectives should include:
   • Understanding onboarding flow and business goals
   • Identifying compliance and audit requirements
   • Discovering operational constraints and manual review processes
   • Clarifying functional and non-functional requirements
   • Exposing potential risks and gaps

3. Generate role-based interview questions with AI
   Use a generative AI tool such as ChatGPT, Claude, or Copilot to create 10 interview questions, grouped by stakeholder role.

Sample prompt:
Generate 10 role-based discovery interview questions for product, compliance, and operations stakeholders for a digital loan onboarding system at an online bank. Focus on business objectives, user experience expectations, compliance requirements, and operational constraints. Group the questions by role.

Part 2: Simulate stakeholder interviews

You will simulate interviews using AI to gather realistic stakeholder responses and uncover assumptions or gaps.

1. Select interview simulation approach
   You will conduct interviews in Q&A format. Each AI response should represent a specific stakeholder role.

2. Run AI interview simulation
   Copy your interview questions from Part 1 and prompt your AI tool to respond as each stakeholder.

Ask the AI to:
• Answer realistically
• Identify unknowns and dependencies
• Note potential policy or system constraints

Sample prompt:
Act as a Compliance Officer at an online bank. Answer the following interview question about digital loan onboarding. Identify any assumptions or missing compliance controls in your response: [insert your question].

Repeat this process for 8–12 total Q&A pairs.

3. Document insights and gaps
   As you review the transcript:
   • Highlight missing requirements
   • Note conflicts or risks
   • Identify unclear business rules

Part 3: Build the requirements pack

You will turn interview insights into structured project requirements.

1. Extract key requirements
   Review the transcript and list functional requirements (FRs) and non-functional requirements (NFRs), such as latency, throughput, availability, RTO/RPO, and data retention.

2. Prioritize requirements using MoSCoW
   Categorize requirements as:
   • Must – KYC, customer consent, soft credit pull, pre-approval decision, audit logging
   • Should – Manual review queue with SLA, email/SMS notifications
   • Could – In-app agent chat, OCR for pre-filled forms
   • Won't – Cross-sell offers, multilingual UI (for MVP)

3. Write user stories
   Convert requirements into 8–12 user stories using this format:
   As a [user], I want [feature], so that [business value].

4. Add acceptance criteria
   For each story, write Given/When/Then acceptance criteria:
   Given that the customer submits identity documents
   When the system validates their identity
   Then the application status updates to "Verified"

Task 1 Deliverables

By the end of this task, you should have the following deliverables:

- **Interview plan:** 10 role-based stakeholder interview questions (Product, Compliance, Operations)
- **Simulated stakeholder interviews:** 8–12 Q&A interview transcript
- **Requirements pack:**
  - Functional and Non-functional requirements identified
  - MoSCoW prioritization list (Must/Should/Could/Won't)
  - 8–12 user stories
  - Given/When/Then acceptance criteria for each user story

You will submit these in one document, Task1_Discovery_and_Requirements, for your final project.

Next steps

You've successfully completed Task 1. Save your Task1_Discovery_and_Requirements document. You'll submit all of your documents after you've completed the entire final project lab.

Next, you'll create a process model and architecture.

Task 2: Create a process model and architecture

In this task, you will use generative AI to model the digital loan onboarding workflow for an online bank. You will create three architecture and process artifacts: A BPMN diagram, a customer journey map, and a component and data flow diagram.

Instructions

Part 1: Create a to-be BPMN process model

In this part, you will create a Business Process Model and Notation (BPMN) diagram that represents the future-state (to-be) onboarding process.

1. Define the process scope
   This process begins when a customer initiates a loan application and ends when the system generates an automated pre-approval decision or routes the case to manual review.

2. Identify key workflow steps
   Before using AI, list the major tasks in the loan onboarding process:
   • Start application
   • Capture personal and KYC information
   • Perform soft credit pull
   • Evaluate credit decision
   • Route to approval or manual review
   • Notify customer

3. Use AI to generate a BPMN
   Use an AI tool such as Mermaid, Lucidchart, Miro, or ChatGPT with Mermaid syntax to generate a BPMN diagram.

Sample AI prompt:
Generate a BPMN model for an online bank's digital loan onboarding workflow. Include: application start, KYC data capture, soft credit pull, decision gateway for credit result, KYC failure path, manual review queue, automated approval, and customer notification. Use standard BPMN notation.

4. Validate branching logic
   Ensure the BPMN includes:
   • A decision gateway for credit score results
   • A KYC verification branch
   • A manual review lane
   • A customer notification end event

Part 2: Design a customer journey map

You will model the customer experience throughout the digital onboarding flow.

1. Define the journey stages
   Your journey map must include these five stages:
   • Stage: Description
   • Discover: Customer sees loan offer online
   • Apply: Customer submits loan application
   • Pre-approve: System evaluates application
   • Verify: Documents/KYC are checked
   • Fund: Approved loan is finalized

2. Identify customer actions and emotions
   Use AI to help describe user expectations, pain points, and emotional state at each stage.

3. Use AI to create a journey map layout
   Ask your AI tool to create a structured journey map in table or diagram format.

Sample AI prompt:
Create a customer journey map for a digital loan onboarding process with five stages:

Discover → Apply → Pre-approve → Verify → Fund

Include customer actions, goals, emotions, and system touchpoints at each stage of the process.

Part 3: Build the component and data flow diagram

You will describe how systems exchange data during onboarding.

1. Identify system components
   Include these components in the diagram:
   • Customer channels (Web and Mobile)
   • API Gateway
   • Credit Bureau System (for soft pull)
   • KYC Service
   • Core Banking System
   • Audit Logging Service

2. Define data flow
   Map the data interactions between services, such as:
   • KYC request → KYC Service → Verification result
   • Credit API call → Credit Bureau → Credit score
   • Application data → Core Banking → Application record

3. Use AI to generate a diagram
   Use Mermaid, draw.io, or Miro and ask AI to generate structured code or diagram steps.

Sample AI prompt:
Generate a system component and data flow diagram for a digital loan onboarding process. Include: Web app, Mobile app, API Gateway, Credit Bureau API, KYC Service, Core Banking database, and Audit Logging. Show direction of data flows.

Task 2 Deliverables

By the end of this task, you should have the following deliverables:

- To-Be BPMN process model
- Customer journey map
- Component and data flow diagram

You will submit these in one document, Task2_Process_and_Architecture, for your final project.

Next steps

You've successfully completed Task 2. Save your Task2_Process_and_Architecture document. You'll submit all of your documents after you've completed the entire final project lab.

Next, you'll define the rules, data, and insights with generative AI.

Task 3: Define the rules, data, and insights with generative AI

In this task, you will define the business rules, data structure, and reporting insights needed for the digital loan onboarding system for an online bank. You will create three deliverables:

- A business rules catalog
- A data entity model
- A reporting dashboard outline

Instructions

Part 1: Define business rules

Define 6–10 business rules for the system. Business rules determine how the loan onboarding process behaves under certain conditions and help ensure consistent decision-making.

1. Identify rule categories
   Your business rules must address:
   • Application completeness checks
   • KYC verification logic
   • Credit eligibility thresholds
   • Retry logic/backoff handling for APIs
   • Decision routing
   • Fraud or risk flags (optional)

2. Use AI to draft business rules
   Ask your AI tool to generate a list of business rules that match the loan processing workflow.

Sample AI prompt:
Generate 8 business rules for an online bank's digital loan onboarding process. Include rules for application completeness, KYC verification, soft credit pulls, minimum credit score thresholds, retry logic for third-party services, and decision routing. Present each rule clearly.

3. Review and refine the rules
   Make sure each rule:
   • Uses clear IF/THEN logic
   • Includes threshold values where needed
   • Covers failure and retry logic
   • Supports system behavior decisions

4. Organize rules in a catalog
   Document each rule with:
   • Rule name
   • Description
   • Trigger condition
   • Expected behavior

   Example template:
   Rule ID: BR-01
   Rule Name: KYC completeness
   Description: Application must include all required identity fields
   Logic: IF KYC data is missing THEN application status = "Incomplete"

Part 2: Define data entities and attributes

You will identify the core data entities used in the onboarding process and define their attributes.

1. Identify key entities
   Your solution must define data entities for:
   • Customer: Applicant information
   • Application: Loan request
   • KYCCheck: Identity verification results
   • CreditPull: Soft or hard credit check
   • AuditLog (optional): Regulatory tracking

2. Use AI to structure entities
   Ask AI to generate attributes for each data entity.

Sample AI prompt:
Create a table of core data entities used in a digital loan onboarding system for an online bank. Include entities Customer, Application, CreditPull, and KYCCheck. For each entity, provide 6–8 attributes with sample descriptions and field types.

3. Validate data relationships
   Make sure entity relationships are clear:
   • Customer submits Application
   • Application triggers KYCCheck
   • KYCCheck stores results
   • Application requests CreditPull

4. Document entity model
   Present the entity definitions in a table format, including field names and descriptions.

Part 3: Define insights and dashboard metrics

You will identify key performance indicators (KPIs) and design a reporting dashboard to monitor onboarding success.

1. Define KPIs
   Select metrics that reflect operational and business performance, such as:
   • Application completion rate
   • Approval rate
   • Average processing time

• KYC failure rate
• Manual review volume

2. Use AI to structure insights
Ask your AI tool for dashboard logic and KPI definitions.

Sample AI prompt:
Generate 8 KPIs for a loan onboarding dashboard. Include calculations for approval rate, conversion rate, average processing time, dropout rate, and KYC failure rate. Explain each KPI.

3. Design a dashboard layout
Sketch a Power BI or Tableau dashboard layout with:
• A funnel chart for application progression
• A bar or pie chart for decision outcomes
• Trendline for weekly approval rate
• KPI summary tiles

Task 3 Deliverables

By the end of this task, you should have the following deliverables:

- **Business rules catalog:** 6–10 business rules
- **Data entities and attributes** defined
- **Insights/dashboard plan:** KPI definitions and dashboard layout description

You will submit these in one document, Task3_Rules_Data_Insights, for your final project.

Next steps

You've successfully completed Task 3. Save your Task3_Rules_Data_Insights document. You'll submit all of your documents after you've completed the entire final project lab.

Next, you'll define quality, deployment, and compliance.

Task 4: Define quality, deployment, and compliance

In this task, you will define how the system will be tested, validated, deployed, and monitored for compliance. You will create quality assurance assets and operational plans using generative AI to support test coverage and system readiness.

Instructions

Part 1: Create test cases

In this part, you will create 6–10 test cases that verify critical functionality of the digital loan onboarding system. Test coverage must include positive, negative, and edge scenarios, and should utilize either Gherkin ("Given/When/Then") or table format.

1. Identify high-priority test scenarios
Based on earlier requirements, include tests for:
• Successful application submission
• KYC validation
• Soft credit pull

• Decision routing
• Application incomplete scenarios
• Error handling or system retry behavior

2. Use AI to generate Gherkin test cases
   Ask AI to create structured test cases based on your user stories.

Sample AI prompt:
Generate 8 test cases for a digital loan onboarding system at an online bank using Gherkin format. Include positive tests for successful onboarding and negative tests for KYC failure, missing data, and credit API timeout.

3. Organize test cases in a table or list format
   Ensure each test case includes:
   • Test ID
   • Description
   • Preconditions
   • Steps (Gherkin)
   • Expected result

Part 2: Build a User Acceptance Testing (UAT) plan

You will outline a UAT plan to validate that the system meets business requirements before release.

1. Define UAT scope
   Describe what UAT will validate:
   • End-to-end onboarding workflow
   • Business rule accuracy
   • Integration reliability
   • User experience expectations

2. Identify roles and responsibilities
   Define who participates in UAT:
   • Product Owner: Approves business functionality
   • QA Lead: Manages test execution
   • Compliance Reviewer: Validates audit and data handling
   • Operations Analyst: Verifies manual review features

3. Establish entry and exit criteria
   Entry criteria examples:
   • Requirements documented
   • UAT environment ready
   • Test data prepared

   Exit criteria examples:
   • 95% of test cases passed
   • No high-severity defects open
   • Compliance checks complete

4. Include defect tracking and SLA
   Define how defects will be managed:
   • Severity levels (Critical/High/Medium/Low)
   • Resolution time SLA
   • Re-test expectations

Part 3: Create a deployment and rollback readiness checklist

You will prepare a deployment readiness plan that ensures the safe launch of the onboarding system and provides rollback options.

1. Create a pre-deployment checklist
   You should include the following:
   • Code reviewed and approved
   • Test cases passed
   • Feature flags configured
   • Production credentials secured
   • Monitoring enabled

2. Add post-deployment smoke tests
   Some examples include:
   • Confirm API Gateway connection
   • Validate credit API requests
   • Test customer login
   • Submit a test loan application

3. Define rollback strategy
   You should describe the following:
   • When rollback is needed
   • Who authorizes rollback
   • Automated vs. manual rollback process
   • Data recovery plan

Task 4 Deliverables

By the end of this task, you should have the following deliverables:

- Test cases: 6–10 Gherkin-style test cases
- User Acceptance Testing (UAT) plan
- Deployment readiness checklist
- Rollback checklist
- Compliance/performance/scalability brief

You will submit these in one document, Task4_Quality_Deployment_Compliance, for your final project.

Next steps

You've successfully completed Task 4. Save your Task4_Quality_Deployment_Compliance document. You'll submit all of your documents after you've completed the entire final project lab.

Next, you'll create communication and platform stubs.

Task 5: Create communication and platform stubs

In this task, you will create professional communication artifacts and simulate platform implementation using AI-assisted tools. This helps demonstrate your ability to communicate business outcomes clearly and structure work within delivery platforms such as Jira, Confluence, and Miro.

Instructions

Part 1: Write an executive brief

Create a one-page Executive Brief that introduces the project to sponsors and leadership stakeholders.

1. Summarize project purpose
   Write a short overview of the problem being solved by the digital loan onboarding system for an online bank.

2. Highlight business value
   Use AI to help articulate benefits such as customer experience improvements, faster loan decisions, and compliance readiness.

3. Describe project scope at a high level
   Summarize what is included in the MVP (KYC, soft credit pull, decision engine) and what is out of scope.

Sample AI prompt:
Write a professional executive summary for a digital loan onboarding system at an online bank. Include problem statement, business goals, key system capabilities, and measurable outcomes. Keep it concise and business-focused.

Part 2: Create a stakeholder map and engagement plan

You will define your stakeholders and outline how you will communicate with them throughout the project.

1. Identify stakeholders
   List relevant teams such as Product, Compliance, Operations, IT, Risk, Lending, and Customer Support.

2. Create a stakeholder map
   Utilize AI to categorize stakeholders based on their influence and interests.

3. Build an engagement plan
   Define communication methods and frequency using a RACI or communication plan table.

Sample AI prompt:
Create a stakeholder engagement plan for a digital loan onboarding project. Include stakeholder roles, responsibilities, communication frequency, and preferred communication channels.

Part 3: Generate a multilingual executive brief

Provide a translated version of your Executive Brief to support global communication.

1. Select a second language
   Choose one language, such as Spanish, French, German, or Hindi.

2. Use AI translation
   Translate the Executive Brief to demonstrate multilingual communication skills.

3. Review tone and accuracy
   Keep language neutral and professional.

Sample AI prompt:
Translate the following executive summary into Spanish using a professional business tone. Retain technical clarity and correct terminology.

Part 4: Create platform implementation stubs

Create lightweight platform artifacts that simulate how this project would be managed in a real delivery environment.

1. Create a Jira Epic and 3 User Stories
   Use AI to generate:
   • 1 Epic to summarize onboarding workflow development
   • 3 User Stories tied to system capabilities (KYC, credit pull, decision engine)

Sample AI prompt:
Create one Jira Epic and three Jira user stories for a digital loan onboarding system. Follow standard Jira formatting. Include acceptance criteria.

2. Create Confluence page outline
   Draft a project page outline, including:
   • Project overview
   • Architecture diagrams
   • Requirements links
   • Risks & assumptions
   • Decision log

3. Generate diagram exports
   If diagrams were created in earlier tasks (BPMN, data flow), simulate including them as attachments or exports to align with real documentation.

Task 5 Deliverables

By the end of this task, you should have the following deliverables:

- Executive brief (1 page)
- Stakeholder map
- Stakeholder engagement plan
- Multilingual executive summary
- Platform stubs:
- Jira epic
- Three Jira user stories
- Confluence page outline
- Diagram exports or links (BPMN/data flow)

You will submit these in one document, Task5_Communication_Platform_Stubs, for your final project.

Next steps

You've successfully completed Task 5. Save your Task5_Communication_Platform_Stubs document. You'll submit all of your documents after you've completed the entire final project lab.

Next, you'll reflect on AI usage.

Task 6: Reflect on your AI usage

In this final task, you will provide a clear, structured reflection on how you used generative AI throughout this project. You will explain what AI supported, how you verified and improved outputs, and how you ensured ethical and responsible use.

Instructions

Part 1: Summarize your AI workflow

Step-by-step, explain exactly how you used generative AI tools during the project.
Your workflow description must include:

1. Which AI tools you used (for example, ChatGPT, Claude, Copilot)

2. The tasks where you applied AI (for example, creating interview questions, generating user stories, drafting diagrams, summarizing stakeholder input)

3. The types of prompts or instructions you used (for example, "Generate BPMN steps for . . . " or "Draft user stories for . . . ")

4. How you reviewed and refined the AI's responses (for example, editing user stories, correcting business rules, adjusting diagrams)

5. Where you relied on your own analysis and reasoning instead of AI

Note: Be specific! Describe what you did and how you did it, not just that AI helped.

Part 2: Document responsible and ethical AI use

Explain how you ensured ethical and responsible AI usage.
You must address:

1. How you avoided exposing private or sensitive data
2. How you verified that AI-generated content was accurate and aligned with banking and compliance requirements
3. How you ensured AI supported — not replaced — your judgment and decision-making
4. How you avoided copying AI content without evaluating or adapting it
5. Any steps you took to remove incorrect, biased, or unrealistic information

Part 3: Describe AI limitations and human review

Provide a short evaluation of AI limitations you encountered and how you handled them.
Include:

1. Where AI answers were incomplete, inaccurate, or generic
2. Any domain-specific knowledge AI did not fully understand (for example, compliance requirements)
3. Examples of adjustments or corrections you made
4. How you applied human judgment to validate content quality and accuracy

Task 6 Deliverables

By the end of this task, you should have the following deliverables:

- **AI usage note:**
  - Reflection on AI use
  - Ethical considerations

- Limitations and human review

You will submit these in one document for your final project.

You've successfully completed Task 6. Save your Task6_AI_Usage_Reflection document. You'll submit all of your documents after you've completed the entire final project lab.

Submitting your project

Organize all deliverables into a comprehensive package with the following structure:

Required files:

1. Task1_Discovery_and_Requirements
2. Task2_Process_and_Architecture
3. Task3_Rules_Data_Insights
4. Task4_Quality_Deployment_Compliance
5. Task5_Communication_Platform_Stubs
6. Task6_AI_Usage_Reflection

You can submit your files as either DOCX (using a word processing software such as Google Docs or Microsoft Word) or PDF.

How to submit your assignment:

Proceed to the **Final Project: Submission and Evaluation** assignment in this course and follow the instructions for submitting your final project. You **MUST** submit these assignments in the Final Project: Submission and Evaluation assignment to successfully finish this project.

Submission format:

- You will be asked to submit all 6 required files in the specified formats one by one in the Final Project: Submission and Evaluation assignment.